

- The sandbag can be positioned lengthways in the intended direction of the carry or positioned across the body. Due to the width of the sandbag, it may be necessary to rotate the hips outward, with the elbows inside the knees.
- Wedge the fingers under the sides of the sandbag so that the hands act as hooks under each side of the sandbag.

Liftoff Position

- After strongly bracing the spine and ensuring a strong grip on the sandbag, forcibly push the feet into the ground and row the sandbag into your chest.
- If the sandbag is quite heavy, transition into a full squat position. However, if the sandbag is light to moderately heavy, it may be possible to bypass the full squat position and, via forceful hip extension, stand up from this position and initiate the carrying phase.

Full Squat Position

- As the sandbag approaches the chest at the end of the liftoff phase, drop the hips rapidly into a deep squat position so that the sandbag is tightly held against the chest and resting on the upper thighs.
- Once secure and balanced in this position, initiate the concentric phase of the squat movement by vigorously extending the hip and knee joints while continuing to strongly pull the sandbag into the chest.
- Arms may be kept on the side of or above the sandbag, depending on personal preference.

Carrying Phase

- From here, adopt a walking movement pattern, initially focusing on performing a series of short, fast steps, which will increase total body balance and stability.
- When building up speed, increase the length of the steps and further reduce ground contact times.

Note: Lean back a little during the walking phase so as not to lose balance forward due to the sandbag positioned in front of the center of mass.

Ending Position

- After completing the set distance, lean slightly forward and drop the sandbag to the floor.

MAJOR MUSCLES INVOLVED

gastrocnemius, soleus, gluteus maximus, rectus femoris, gluteus medius, erector spinae, internal and external obliques



Starting position



Liftoff position



Full squat recovery



Carrying phase

13 ZERCHER CARRY

Starting Position and Liftoff

The Zercher carry can begin from one of two positions.

1. Starting position from boxes

- Position the barbell on some boxes, anywhere from a full squat position to standing almost completely upright.
- Position the arms under and around the bar with the palms facing up so that the bar now rests on the inner part of the elbows and is held closer the upper abdominal region. It might be easier to maintain this position by clasping the hands and keeping them close to the chest (not shown).

2. Starting position from the floor

- Position the barbell on the floor near the shins with a closed overhand grip and then perform a semi-sumo squat deadlift.
- Once the bar is above the knees, drop into a full semi-sumo squat position and reposition the arms under and through the bar so that the bar is on the inner part of the elbows and contacting the upper abdominal region.

While maintaining the position of the bar on the inner elbows and upper abdominals, forcibly push the feet into the ground and extend the knees and hips, now standing upright with the bar kept in the same relative position as liftoff.

Walking Phase

- Adopt a walking movement pattern, initially focusing on performing a series of short, relatively fast steps, which will increase total body balance and stability.
- When building up speed, increase the length of the steps and further reduce ground contact times.

Note: Lean back a little during the walking phase so as not to lose balance forward due to the barbell positioned in front of the center of mass.

Ending Position

- After completing the set distance, reset the bar on the boxes by performing the eccentric phase of a full or partial squatting motion, depending on the height.

- Alternatively, if repositioning the bar on the floor, go into a full squat position, and while resting the bar on the upper thighs, rotate the arms out from under the bar so that the hands reach the top of the bar.
- From this position, grab the bar with an overhand grip to partially stand upright before replacing the bar on the ground by performing the concentric phase of a deadlift-type motion.

MAJOR MUSCLES INVOLVED

gastrocnemius, soleus, gluteus maximus, rectus femoris, gluteus medius, erector spinae, internal and external obliques



Starting position from boxes



Starting position from floor



Liftoff from floor (mid position)



Liftoff from floor (standing position)



Walking phase

14 FORWARD SLED PUSH AND PULL

Starting Position (Push)

- Stand behind the sled and position the hands on the vertical bars of the prowler at a comfortable height, typically somewhere between hip and knee height in a standing position. (*Note:* The lower the hands are positioned on the prowler, the more likely this will result in a horizontal trunk and shin position, which will increase the ability to apply propulsive horizontal forces and accelerate the sled.)
- Keep the hands closer to the shoulders and elbows near the hips (shown) or extend the arms forward in front of the body (not shown), depending on personal preference.

- If the load is heavy, it is advisable to start with the two feet positioned together (not shown) to maximize the ability to apply horizontal force to initiate movement of the heavy sled.
- If the load is lighter, it is possible to start in an alternate split stance (shown) to better replicate a sprint starting position.

Forward Pushing Phase

- While bracing the spine, initiate a stepping pattern that initially focuses on performing a series of short, relatively fast steps, which will increase the ability to apply horizontal forces to the ground.
- While building up speed, increase the length of the steps and further reduce ground contact times. The ability to apply propulsive horizontal forces to the ground can be optimized by ensuring the front foot is positioned behind the hip throughout the entire pushing phase.

Starting Position (Pull)

- Stand in front of the sled and attach the sled to the body via a chest and shoulder harness (shown) or hold separate handles in each hand (not shown).
- Regardless of the attachment method to the sled, assume a position in which the trunk and shin are close to horizontal (not shown), which will increase the ability to apply propulsive horizontal forces and accelerate the sled. This can be best achieved by placing the hands on the ground (not shown) if connected via a harness to the sled or by leaning forward (not shown) if holding handles in each hand.
- If the load is heavy, it is advisable to start with the two feet positioned together (not shown) to maximize the ability to apply horizontal force to initiate movement of the heavy sled. If the load is lighter, it is possible to start in an alternate split stance (shown) to better replicate a sprint starting position.

Forward Pulling Phase

- While bracing the spine, initiate a stepping pattern that initially focuses on performing a series of short, relatively fast steps, which will increase the ability to apply horizontal forces to the ground.
- While building up speed, increase the length of the steps and further reduce ground contact times. The ability to apply propulsive horizontal forces to the ground can be optimized by ensuring the front foot is positioned behind the hip throughout the entire pushing phase.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, gastrocnemius, soleus



Starting position (push)



Forward pushing phase



Starting position (pull)



Forward pulling phase

15

REVERSE SLED DRAG

Starting Position

- Stand in front of the sled and attach the sled to the body via a hip harness (not shown) or hold separate handles in each hand (shown).
- Regardless of the attachment method to the sled, lean backward a little while maintaining a relatively extended hip position, and keep the feet in front of the knees to increase the ability to apply propulsive horizontal forces and accelerate the sled.
- It is advisable to start with the two feet positioned together to maximize the ability to apply horizontal force to initiate movement of the heavy sled.

Backward Pulling Phase

- Initiate a stepping pattern that initially focuses on performing a series of short, relatively fast steps, which will increase the ability to apply horizontal forces to the ground.
- While building up speed, increase the length of the steps and further reduce ground contact times.
- The ability to apply propulsive horizontal forces to the ground can be optimized by ensuring the front foot is typically positioned in front of the knee throughout the entire pulling phase.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, gastrocnemius, soleus



Starting position



Backward pulling phase

Other Alternative Exercises

16 BACK SQUAT WITH BANDS

Band Placement

- Loop the ends of long elastic bands around the band pegs of each side of a power rack. If the rack does not have band pegs,

loop the bands around the handles of heavy dumbbells that will not move during the exercise.

- Take the other ends of the bands and place them over the ends of the bar so they rest against the outside of the weight plates on either side.
- The bands should be loose enough that there is no tension at the bottom of the squat. Tension should be applied to the bar when the exercise is performed, but at the bottom of the range of motion, the bands should be slack to apply no load to the bar. (*Note:* Despite their appearance, the bands in the second photo are not creating any tension on the bar.)

Starting Position

- Grasp the bar with a closed, pronated grip.
- Step under the bar and position the feet parallel to each other.
- Place the bar in a balanced position on the upper back and shoulders; follow the guidelines for the back squat in [chapter 16](#) regarding the high or low placement of the bar.
- Extend the hips and knees to lift the bar off the supporting pins or ledge, and take one or two steps backward.
- Position the feet shoulder-width apart or wider, even with each other, and the toes pointed slightly outward.
- Stand with an erect torso by positioning the shoulders back, tilting the head slightly back, and protruding the chest up and out to create a neutral or slightly arched back.
- All repetitions begin from this position.

Downward Movement Phase

- Begin the exercise by flexing the hips and knees slowly and under control while keeping the torso-to-floor angle relatively constant.
- Maintain a flat-foot position with the knees aligned over the feet as they flex.

- Continue flexing the hips and knees until the thighs are parallel to the floor, the torso begins to round or flex forward, or the heels begin to rise off the floor.

Upward Movement Phase

- Raise the bar by extending the hips and knees while keeping the torso-to-floor angle relatively constant.
- Keep the torso erect, shoulders back, and chest up and out.
- Keep the heels on the floor and the knees aligned over the feet.
- Continue extending the hips and knees until the body is fully erect.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris



Starting position



Downward and upward movements

17 ACCENTUATED ECCENTRICALLY LOADED BENCH PRESS

Note: This exercise requires the use of weight releasers, which allow additional load to be applied during the eccentric phase of the movement. The spotting technique used is identical to that outlined for the bench press in [chapter 16](#).

Set-up

- Prior to the athlete initiating the movement the spotter should load the barbell with the appropriate weight and the weight releasers are attached to the barbell.
- It is important that the weight releasers are loaded correctly and placed evenly on the barbell so that they release at the completion of the downward movement phase without disrupting the athlete's normal movement pattern. As a general rule, the weight releasers should be set so that they disengage from the barbell approximately 2 to 4 inches (5-10 cm) before the completion of the downward movement (51).

Starting Position

- Lie in a supine position on a bench in the five-point body contact position.
- Place the body on the bench so that the eyes are below the racked barbell.
- Grasp the barbell with a closed, pronated grip slightly wider than shoulder-width apart.
- Signal the spotter for assistance in moving the barbell off the supports.
- Position the barbell over the chest with the elbows fully extended.
- All repetitions begin from this position.

Downward Movement Phase

- Lower the barbell under control to reduce the movement of the weight releasers.
- The weight releasers will disengage from the barbell about 2 to 4 inches (5-10 cm) before touching the chest at approximately nipple level.
- Keep the wrists stiff and the forearms perpendicular to the floor and parallel to each other.
- Maintain the five-point body contact position.

Upward Movement Phase

- Push the barbell upward and very slightly backward as fast as possible until the elbows are fully extended.
- Keep the wrists stiff and the forearms perpendicular to the floor and parallel to each other.
- Maintain the five-point body contact position.
- Do not arch the back or raise the chest to meet the bar.
- At the end of the set, signal the spotter for assistance in racking the bar.
- Keep a grip on the bar until it is racked.

MAJOR MUSCLES INVOLVED

pectoralis major, anterior deltoids, triceps brachii

18 LANDMINE ROTATION

Starting Position

- Stand facing the free end of the barbell, then squat down to grasp the end of the barbell sleeve with both hands (not shown) and pick it up.
- After standing, press the free end of the barbell up to position it in front of the midline of the body approximately level with the eyes.
- Stand with the feet approximately shoulder-width apart and knees slightly flexed.

- The body weight should be shifted to the balls of the feet but with the heels remaining in contact with the floor.
- Brace the trunk with the head and spine in a neutral position (4).

Rotate to the Left

- Lower the bar by rotating to the left while also allowing the torso, hips, and back foot (i.e., the foot opposite the direction of the bar rotation) to rotate in the same direction of the barbell (4) (somewhat more than what is shown).

Return to the Starting Position

- Using a forceful and rapid motion, rotate the bar back to the starting position while keeping the torso and hips aligned with a neutral head position.
- During this motion keep the arms extended and in front of the torso throughout the rotational elevation of the barbell while returning to the starting position (4).

Rotate to the Right

- Lower the bar by rotating to the right while also allowing the torso, hips, and back foot (i.e., the foot opposite the direction of the bar rotation) to rotate in the same direction of the barbell (4) (somewhat more than what is shown).

Return to the Starting Position

- Using a forceful and rapid motion, rotate the bar back to the starting position while keeping the torso and hips aligned with a neutral head position.
- During this motion keep the arms extended and in front of the torso throughout the rotational elevation of the barbell while returning to the starting position (4).

MAJOR MUSCLES INVOLVED

rectus abdominis, obliques, erector spinae



Starting position



Rotate to the left



Rotate to the right

19

SINGLE-ARM LANDMINE ROW

Setup and Starting Position

- While standing facing away from the axis of rotation of the landmine barbell, assume a hip-width stance with the unfixed side of the barbell just outside of the working-side leg with the toes behind the barbell sleeve.
- While maintaining a 25- to 45-degree knee angle, hinge at the hips until able to grasp the end of the barbell with one hand while the elbow is extended and reaching downward. In this position the forward trunk inclination should be roughly 30 to 60 degrees relative to the ground. This trunk angle should remain constant throughout the movement (80).

Upward Movement Phase

- The upward movement phase (i.e., concentric phase) is initiated from the starting position and ends at the top position where the barbell reaches the greatest angle relative to the floor and contacts the torso.
- Throughout this motion remain stationary with an approximately neutral axial alignment while simultaneously retracting the shoulder girdle, extending the glenohumeral joint, and flexing the elbow to pull the barbell along the natural arc of the landmine barbell device (80).

Downward Movement Phase

- Allow the shoulder girdle to protract, the glenohumeral joint to flex, and the elbow to extend simultaneously to return to the starting position (80).

Note: As part of a balanced training program, this movement is performed on both sides of the body.

MAJOR MUSCLES INVOLVED

latissimus dorsi, trapezius, teres major, posterior deltoid, triceps brachii, rhomboids, brachioradialis, biceps brachii, brachialis



Setup, starting, and ending positions



End of upward movement phase

20

LANDMINE SQUAT TO PRESS

Setup and Starting Position

- Stand facing the axis of rotation of the landmine barbell with a slight forward lean, while assuming a shoulder-width stance with the knees slightly flexed and grasping the end of the bar with one hand.
- The bar should be positioned about 4 to 6 inches (10.2-15.2 cm) away from the shoulder of the arm that is holding the bar (135). The other arm is held out to the side of the body and used for balance during the exercise.

Downward Movement Phase

- Begin the exercise by flexing the hips and knees slowly and under control while keeping the torso-to-floor angle relatively constant.
- Maintain a flat-foot position with the knees aligned over the feet as they flex.
- Continue flexing the hips and knees until the thighs are parallel to the floor, the torso begins to round or flex forward, or the heels begin to rise off the floor.

Upward Movement Phase

- Extend the hips and knees while keeping the torso-to-floor angle relatively constant.
- Keep the torso erect, shoulders back, and chest up and out.
- Keep the heels on the floor and the knees aligned over the feet.
- Continue extending the hips and knees; as the fully-standing position is reached, press the bar up and slightly forward until the elbow of the arm holding the bar is fully extended; full knee and elbow extension should be attained at about the same time.

Return to the Starting Position

- Keeping the lower body and torso stationary, slowly lower the bar in a controlled manner until it is returned to the starting position.

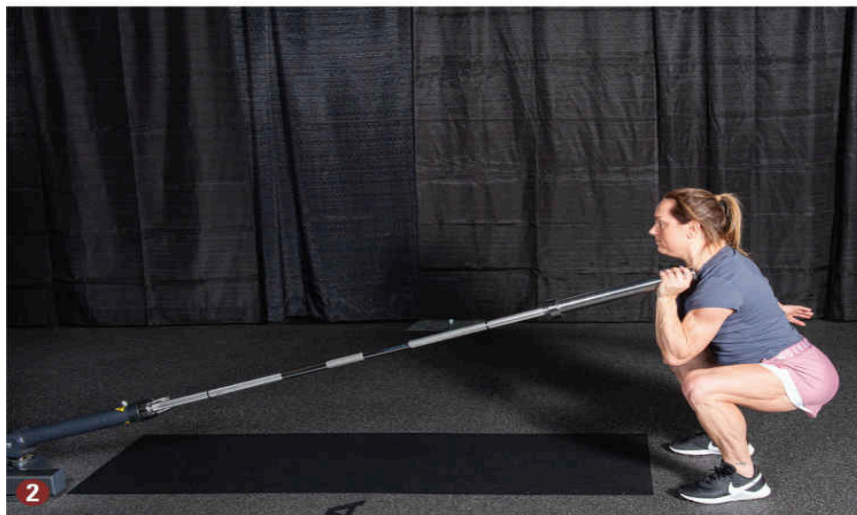
Note: As part of a balanced training program, this movement is performed on both sides of the body.

MAJOR MUSCLES INVOLVED

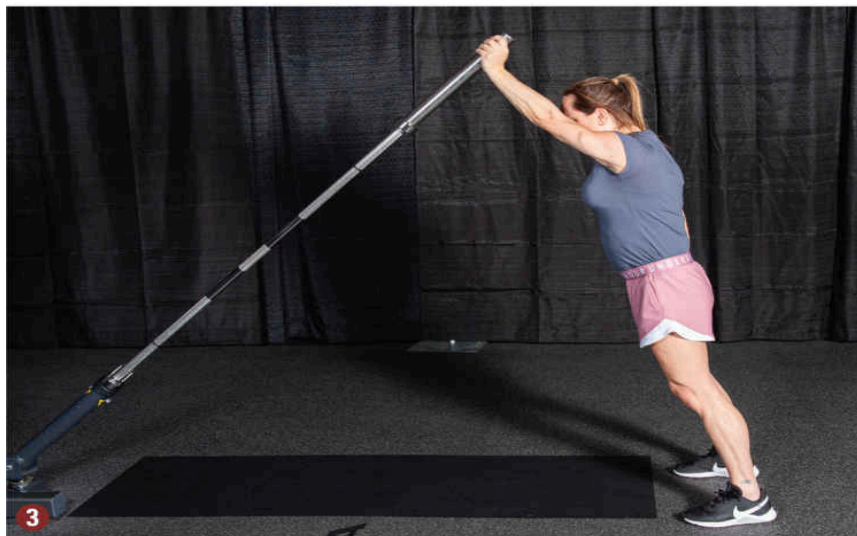
gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, deltoid (anterior, medial, and posterior), trapezius, triceps brachii



Setup, starting, and ending positions



End of downward movement phase



End of upward movement phase

21

TWO-ARM KETTLEBELL SWING

Starting Position

- Stand with feet flat and placed between hip- and shoulder-width apart with the toes pointed straight ahead, straddling a kettlebell.
- Squat down with the hips lower than the shoulders and grasp the kettlebell with a closed, pronated grip.
- Place the hands on the kettlebell with index fingers touching or close together, inside of the legs, keeping the elbows fully extended.
- Position the body with the back in a neutral spine position, shoulders retracted and depressed, feet flat on the floor, and eyes focused straight ahead or slightly upward.
- Maintain a neutral spine or normal lordotic position while flexing the hips and knees to approximately a quarter-squat position, with the kettlebell hanging at arm's length between the thighs.
- All repetitions begin from this position.

Backward Movement Phase

- Begin the exercise by flexing at the hips to swing the kettlebell between the legs.
- Keep the knees in a moderately flexed position with the back neutral and the elbows extended.
- Keep swinging the kettlebell backward until the torso is nearly parallel to the floor and the kettlebell is past the vertical line of the body.

Forward and Upward Movement Phase

- When the backward swing reaches its end point, reverse the movement by extending the hips and knees to move the kettlebell in an upward arc.

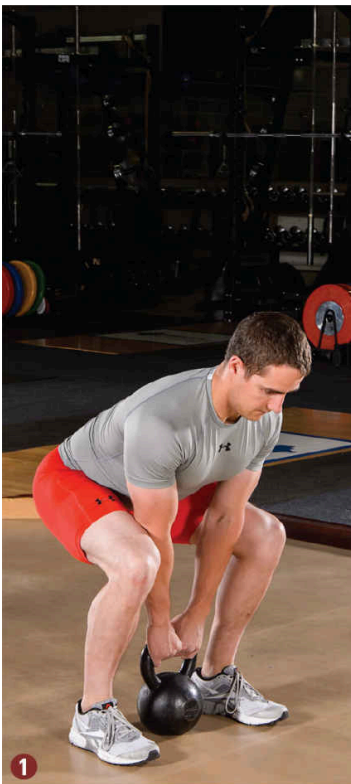
- Allow momentum to raise the kettlebell to eye level, keeping the elbows extended and the back in a neutral spine position.

Downward and Backward Movement Phase

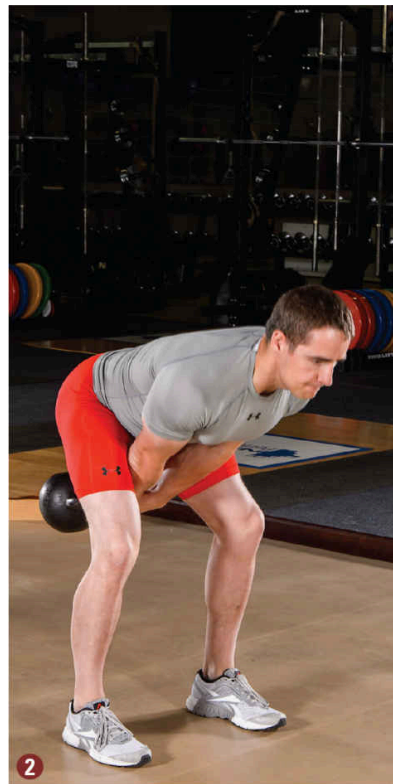
- Allow the kettlebell to drop into the downswing; flex the hips and knees to absorb the weight. Keep the elbows fully extended and the back neutral.
- Continue the downward-then-backward movement until the kettlebell passes under then behind the body, then begin the upward movement phase for the next repetition.

MAJOR MUSCLES INVOLVED

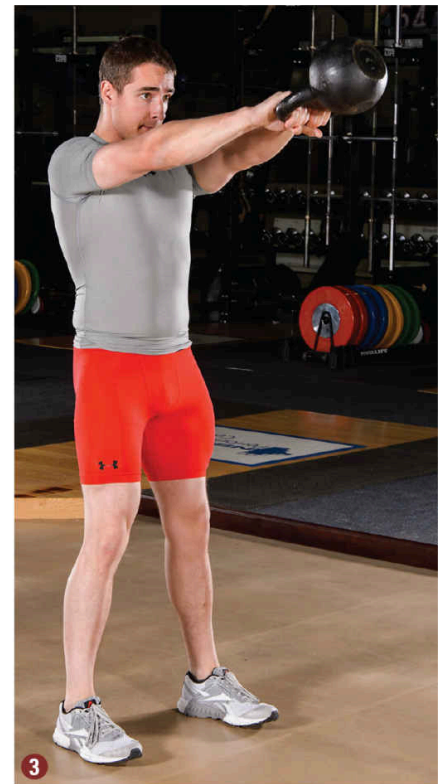
gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris



Starting position



Initial backward movement and beginning of forward and upward movement



End of forward and upward movement and beginning of downward and backward movement

22

SINGLE-LEG SQUAT

Note: The single-leg squat is not typically done with a spotter. If spotting is needed, two spotters are used with one on each side of the bar. The positions of the spotters are not pictured in order to show the athlete's position at the various stages of the single-leg squat more clearly.

Starting Position: Athlete

- Grasp a barbell with a closed, pronated grip (for dumbbell variation, hold dumbbells with a closed, neutral grip with a dumbbell in each hand).
- Stand in front of a bench or box that is approximately knee height with the feet approximately shoulder- to hip-width apart.
- Facing away from the bench or box, take a moderate step forward with one leg and place the instep of the back foot on top of the bench or box.
- Allow both knees to be slightly flexed, with the torso in a nearly erect position and the shoulders held back and chest up and out.
- All repetitions begin from this position.

Starting Position: Two Spotters

- The spotters should stand erect at opposite ends of the bar with the feet hip-width apart and the knees slightly flexed.
- Hold the hands 2 to 3 inches (5-8 cm) below the end of the bar.

Note: If dumbbells are used, no spotter is needed.

Downward Movement Phase: Athlete

- Flex the hip and knee of the forward leg simultaneously to lower the body in a vertical plane while keeping the torso-to-floor angle constant.
- Keep the heel of the forward foot flat on the floor and the instep of the back foot on top of the bench or box.

- Continue flexing the hip and knee until the front thigh is approximately parallel to the floor.

Downward Movement Phase: Two Spotters

- Keep the cupped hands close to but not touching the bar as it descends.
- Slightly flex the knees, hips, and torso to keep a neutral spine position when following the bar.

Upward Movement Phase: Athlete

- Raise the bar under control by actively extending the forward hip and knee; extend the other hip and knee also to keep the torso-to-floor angle constant.
- Maintain a neutral spine position and keep the torso upright.
- Keep the forward knee aligned over the forward foot.
- Do not flex the torso forward or round the spine.
- Continue extending the hip and knee of the forward leg to reach the starting position.
- Repeat for the desired number of repetitions and then switch the forward leg.

Upward Movement Phase: Two Spotters

- Keep cupped hands close to but not touching the bar as it ascends.
- Slightly extend the knees, hips, and torso and keep the back neutral when following the bar.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris



Starting position



Downward and upward movements

23 SINGLE-LEG ROMANIAN DEADLIFT (RDL)

Note: This exercise can be performed with the weight held in the hand of the same side of the body as the support leg (ipsilateral) or in the hand of the opposite side of the body as the support leg (contralateral). The description and the photos describe the contralateral single-leg RDL.

Starting Position

- With the right hand, grasp a dumbbell or kettlebell with a closed, pronated grip.
- Stand on the left leg (as the support leg) with the hips and shoulders over the left foot.
- Hold the dumbbell or kettlebell in front of the right thigh with the right elbow fully extended and the right foot slightly staggered back.
- All repetitions begin from this position.

Downward Movement Phase

- Allow the left support knee to flex until it reaches a moderately flexed position, and then rigidly hold that position throughout the movement.
- Begin the exercise by allowing the torso to flex forward at the hip of the left support leg.
- Keep the right shoulder, hip, knee, and ankle in one line as the torso flexes forward.
- Keep the back neutral and the right elbow still fully extended.
- Lower the dumbbell or kettlebell until the torso and right leg are approximately parallel to the floor.

Upward Movement Phase

- Extend the hip of the left support leg to return to the starting position.
- Do not hyperextend the torso or flex the elbow holding the dumbbell or kettlebell.
- Repeat for the desired number of repetitions and then switch the support leg.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris



Starting position



Downward and upward movements

24 ONE-ARM DUMBBELL SNATCH

Starting Position

- Straddle a dumbbell and place the feet between hip- and shoulder-width apart with the toes pointed slightly outward.
- Squat down with the hips lower than the shoulders; grasp the dumbbell with a closed, pronated grip and the elbow fully extended.
- All repetitions begin with the back neutral or slightly arched, scapulae depressed and retracted, and chest held up and out. The head is in line with the vertebral column or slightly hyperextended and the body's weight balanced between the

middle and balls of the feet with feet flat on the floor. The eyes are focused straight ahead or slightly upward. The shoulders of the hand that is holding the dumbbell should be placed over or slightly in front of the dumbbell.

Upward Movement Phase

- Begin the exercise by extending the knees, hips, and ankles forcefully to accelerate the dumbbell.
- The dumbbell should slide up the thigh or remain very close to the thigh as it accelerates upward.
- Keep the elbow of the arm holding the dumbbell fully extended as the knees, hips, and ankles are extending.
- Once the knees, hips, and ankles reach full extension, rapidly shrug the shoulder of the arm holding the dumbbell.
- As the shoulder reaches its highest elevation, flex the elbow holding the dumbbell and keep it close to the torso.
- Keep the arm that is not holding the dumbbell on the opposite hip or held to the side.
- Continue pulling the dumbbell as high as possible.

Catch Phase

- Pull the body under the dumbbell by rotating the arm and hand holding the dumbbell around then under the dumbbell and by flexing the hips and knees to approximately a quarter-squat position.
- Once the arm holding the dumbbell is under the dumbbell, extend the elbow quickly to push the dumbbell up and the body downward under the dumbbell.
- Catch the dumbbell in full extension at the same time the body reaches the quarter-squat position.
- Keep the arm that is not holding the dumbbell on the opposite hip or held to the side.
- After gaining control and balance, stand up to a fully erect position.

Downward Movement Phase

- Slowly allow the dumbbell to lower to the shoulder, then the thigh, and finally to the floor between the feet using a squatting movement.

MAJOR MUSCLES INVOLVED

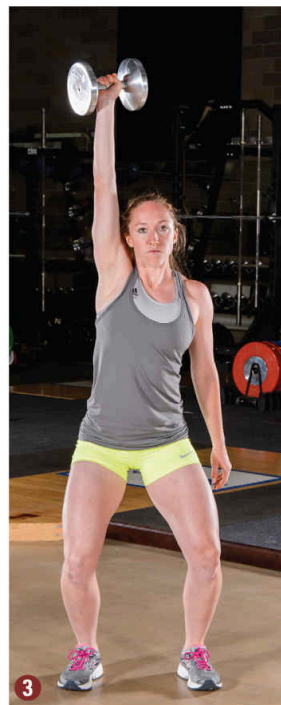
rectus femoris, vastus medialis, vastus lateralis, vastus intermedius, gluteus maximus, semimembranosus, semitendinosus, biceps femoris, deltoids, trapezius



Starting position



Upward movement



Catch



Standing position

KEY TERMS

accentuated eccentric loading (AEL)
accommodating resistance
alternative modes
anatomical core
anterior cruciate ligament (ACL)
augmented eccentric loading

axial skeleton
bilateral asymmetries
bilateral deficit
bilateral facilitation
bodyweight training
chains
constant external resistance
core
farmer's walk
ground-based free weight exercises
isolation exercises
kettlebell
landmine
log lift
logs
machine-based training
muscle activation
nontraditional implement
resistance bands
sandbag carries
sticking point
strongman training
suspension training
unilateral carries
Valsalva maneuver
variable resistance
weighted carries
yoke walk

STUDY QUESTIONS

1. What is the most common injury associated with tire flipping?
 - a. rotator cuff tear
 - b. biceps tear
 - c. lower back strain

- d. hamstring strain
2. Which of the following statements about kettlebell training is most accurate?
 - a. Kettlebell training is proven to be more effective than traditional weightlifting for strength development.
 - b. There is extensive research demonstrating the efficacy of kettlebell training as a strength development tool.
 - c. Most of the scientific evidence supporting the use of kettlebells highlights their potential usefulness for general physical or fitness development.
 - d. Kettlebell training is not appropriate for clinical or recreationally trained populations.
 3. What is a potential solution to the limitation of resistance in suspension training?
 - a. wearing a weighted vest
 - b. using multiple anchor points
 - c. performing exercises on one leg instead of two
 - d. incorporating plyometric movements
 4. What is the recommended percentage of body mass for AEL when trying to enhance performance during the concentric phase of a plyometric jumping exercise?
 - a. 5%-10%
 - b. 10%-20%
 - c. 20%-30%
 - d. 30%-40%
 5. What is bilateral facilitation?
 - a. the tendency for stronger individuals to perform better in sports that require unilateral movements
 - b. an increase in voluntary activation of the agonist muscle group during bilateral movements
 - c. the use of resistance training equipment that allows for both unilateral and bilateral exercises

- d. an injury rehabilitation protocol that focuses on strengthening both limbs equally